

HydroGrid

Seasonal Green Hydrogen Storage for Alpine Independence

A closed-loop energy system for off-grid alpine infrastructure

Off-grid power fails exactly when it's needed most

The Li-ion & Diesel Deadlock

Cold-weather failure Li-ion batteries lose 50%+ efficiency below 0°C, and often fail outright on sub-zero alpine nights.

Diesel dependency Remote stations fall back on fuel hauled over icy, sometimes impassable mountain roads.

Environmental impact Generator noise and emissions disrupt fragile, often protected ecosystems.

The seasonal gap Summer solar surplus goes unused, while winter leaves the same site powerless.

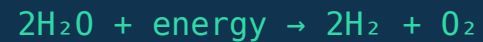
The Closed-Loop Cycle

Turning summer sun into winter power — with pure water as the only byproduct

1

Production

Summer solar surplus drives a PEM electrolyzer, splitting water into H₂ and O₂.



2

Storage

H₂ binds chemically into a solid metal-hydride lattice at low, safe pressure (<30 bar).

3

Generation

In winter, a fuel cell recombines stored H₂ with air, producing electricity, heat, and water.

Why Not Simpler Hydrogen Storage?

Compressed and liquid H₂ both fail the off-grid, unattended-for-months requirement

Compressed gas (350–700 bar)

High explosive risk at unmonitored remote sites, and requires expensive pressure certification.

Liquid hydrogen (–253°C)

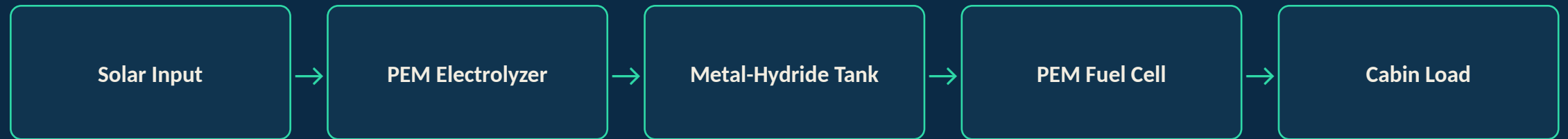
"Boil-off" losses are too high for seasonal storage — the entire reserve can vanish within weeks.

Metal-hydride (HydroGrid's choice)

Near-atmospheric pressure, no boil-off, and stable across the full alpine temperature swing.

System Architecture

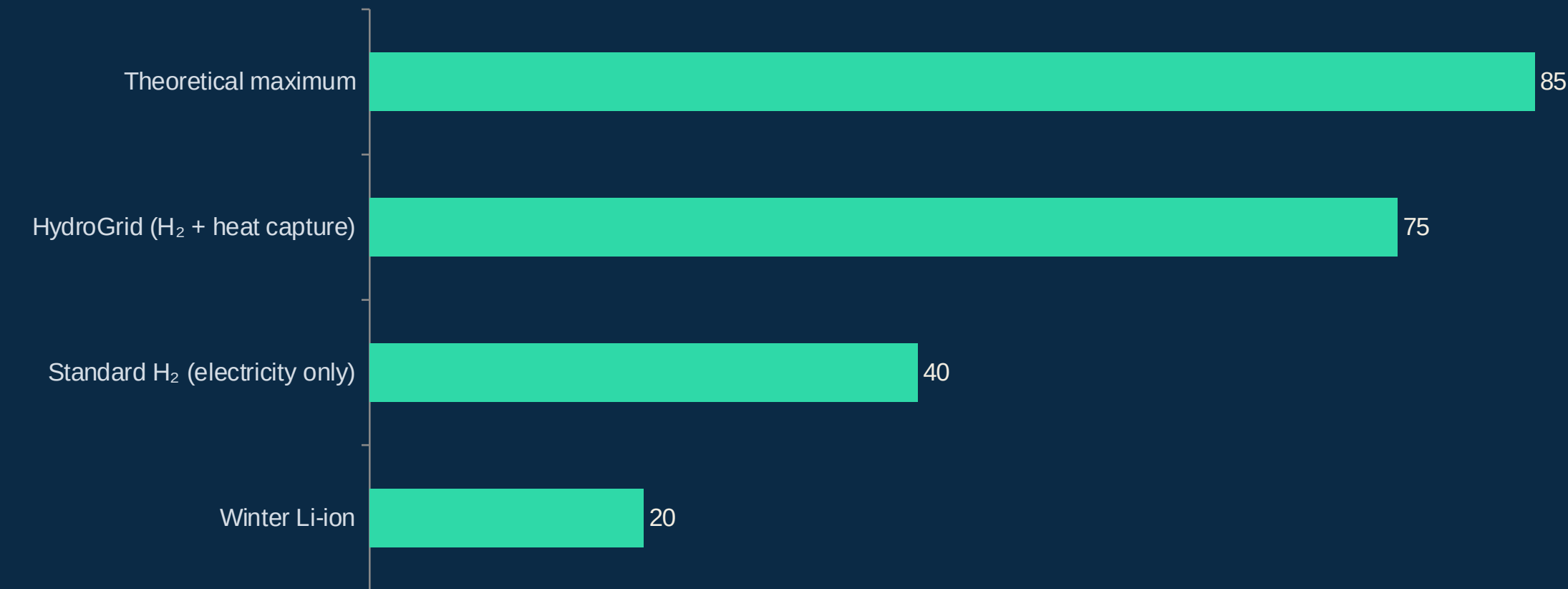
A C++ state-machine firmware layer coordinates every stage of the cycle



- Monitors pressure, temperature, and solar yield in real time
- Automates transitions between charging, storing, and discharging modes
- Enforces safety cutoffs (over-pressure, over-temperature, low-fuel warning)

System Efficiency, End to End

Capturing fuel-cell waste heat for cabin space heating is what pushes overall efficiency to 75%



What That Efficiency Buys You

11

kWh electricity
per storage unit

From ~7,000 liters of stored hydrogen

A single storage cartridge sized like this can power critical instrumentation and radio comms for 4–5 days of complete winter darkness — the exact window when a diesel resupply run would be least possible.

Development Roadmap

Phase 1



Simulation & control-logic development
in C++

Phase 2



Bench-scale prototyping with small
hydride cartridges

Phase 3



Alpine field pilot at a real ranger cabin

Phase 4



Full-scale deployment and regional
scaling

Toward an independent, green alpine future

HydroGrid — a student-led feasibility project

github.com/prostogio/HydroGrid